

An exact column–row phase transition on the row–column interpolation scale

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Abstract

Ray and Sarkar ask which operator spaces have the column–row property. For the canonical Hilbertian interpolation scale $H_\theta = (\mathbf{R}, \mathbf{C})_\theta$ ($0 \leq \theta \leq 1$), we compute every finite column-to-row norm exactly:

$$\gamma_n(H_\theta) = n^{\max\{(1-2\theta)/2, 0\}}.$$

Consequently, the infinite-dimensional space H_θ has the column–row property if and only if $\theta \geq \frac{1}{2}$, and in that range its optimal constant is 1. The reverse norm is $n^{\max\{(2\theta-1)/2, 0\}}$. We also record that the property, with the expected geometric-mean bound on constants, is stable under complex interpolation. This is a complete result for this natural one-parameter family, not a classification of all operator spaces.

1 Source question and transcription

an important tool in extending classical results on Hardy spaces to complete Nevanlinna-Pick (cnp) spaces. In a remarkable recent work [8], Hartz showed that every normalized cnp space has the column-row property with constant 1. The notion of column-row property has led to a plethora of important results for cnp spaces, to name a few: (a) factorization for weak-product spaces; (b) interpolating sequences; (c) Corona problem etc. (see [1, 2, 5, 8, 24]). Motivated by the inherent operator space structure of multiplier algebras and the immense application of this property, Hartz asked the following question in [8]:

Question. *Which operator spaces satisfy the column-row property?*

Our aim in this article is to initiate a study for this question in the general setting of operator spaces, by looking at several examples. For the theory of operator spaces and related important results we refer to the excellent monographs [6, 13, 16]. Let us begin with the description of column-row property by considering a *concrete* operator space $E \subseteq B(\mathcal{H})$. For a sequence $\mathbf{e} := (e_1, \dots, e_n)$ in E , we define the column operator $C_{\mathbf{e}} : \mathcal{H} \rightarrow \mathcal{H}^n$ by $C_{\mathbf{e}}(\zeta) := [e_1(\zeta), \dots, e_n(\zeta)]$, $\zeta \in \mathcal{H}$. Similarly, let us define the corresponding row operator $R_{\mathbf{e}} : \mathcal{H}^n \rightarrow \mathcal{H}$ by $R_{\mathbf{e}}([\zeta_1, \dots, \zeta_n]) := \sum_{j=1}^n e_j(\zeta_j)$.

Definition 1.1. A concrete operator space $E \subseteq B(\mathcal{H})$ is said to have the *column-row property* (in short CRP) if there exists a constant $C > 0$ such that for any finite sequence $\mathbf{e}$ in E with $\|C_{\mathbf{e}}\|_{\mathcal{H} \rightarrow \mathcal{H}^n} \leq 1$ we have

$$\|R_{\mathbf{e}}\|_{\mathcal{H}^n \rightarrow \mathcal{H}} \leq C.$$

The displayed crop is from page 1 of Ray–Sarkar, arXiv:2210.07223v2. The source asks:

Which operator spaces satisfy the column–row property?

For a concrete operator space $E \subseteq B(\mathcal{H})$, the source calls E column–row if one constant K satisfies

$$\|[x_1 \ \cdots \ x_n]\|_{M_{1,n}(E)} \leq K \|[x_1 \ \cdots \ x_n]^t\|_{M_{n,1}(E)} \quad (n \geq 1)$$

for all finite families in E . The paper proves classifications for C^* -algebras and semifinite noncommutative L_p -spaces and notes that **C** and **OH** have constant 1, whereas **R** fails the property. It does not analyze the interpolation continuum joining these three spaces.

2 Definitions and result

Let **R** and **C** be the row and column operator-space structures on ℓ_2 . We fix the convention

$$H_\theta = (\mathbf{R}, \mathbf{C})_\theta, \quad 0 \leq \theta \leq 1,$$

for complex interpolation of operator spaces. Thus $H_0 = \mathbf{R}$, $H_{1/2} = \mathbf{OH}$, and $H_1 = \mathbf{C}$, completely isometrically.

Definition 1. For an operator space E and $n \geq 1$, put

$$\gamma_n(E) = \|t_n : M_{n,1}(E) \longrightarrow M_{1,n}(E)\|,$$

where $t_n([x_1 \ \cdots \ x_n]^t) = [x_1 \ \cdots \ x_n]$. Thus E has the column–row property exactly when $\sup_n \gamma_n(E) < \infty$.

Theorem 2 (Exact phase transition). *For every $0 \leq \theta \leq 1$ and every $n \geq 1$,*

$$\gamma_n(H_\theta) = n^{\frac{1}{2}(1-2\theta)_+}, \quad (s)_+ = \max\{s, 0\}.$$

Hence H_θ has the column–row property if and only if $\theta \geq \frac{1}{2}$. Whenever it has the property, the optimal uniform constant is 1.

Corollary 3 (Reverse direction). *If $\rho_n(E) = \|t_n : M_{1,n}(E) \rightarrow M_{n,1}(E)\|$, then*

$$\rho_n(H_\theta) = n^{\frac{1}{2}(2\theta-1)_+}.$$

In particular, both directions are uniformly bounded only at the symmetric point $H_{1/2} = \mathbf{OH}$.

3 Idea and proof intuition

There are two complementary parts.

First, interpolation gives the upper bound. At the row endpoint the column-to-row transpose has norm $\sqrt{n}$; at the symmetric midpoint **OH** it is an isometry; and at the column endpoint it is a contraction. Reiteration then gives exactly the exponent in Theorem 2.

Second, the first n coordinate vectors detect the bound below the midpoint. Their column and row norms in H_θ are exactly $n^{\theta/2}$ and $n^{(1-\theta)/2}$. The upper estimates follow by interpolating their endpoint norms. A trace-type functional attains the matching lower estimates. This dual witness is what turns a soft interpolation bound into an exact formula.

4 Proof

We use only standard exactness and reiteration for complex interpolation of operator spaces. In particular,

$$M_{a,b}((E_0, E_1)_s) = (M_{a,b}(E_0), M_{a,b}(E_1))_s \quad (4.1)$$

isometrically.

Endpoint norms

For every concrete operator space $E \subseteq B(\mathcal{K})$,

$$\|[x_1 \ \cdots \ x_n]\|^2 = \left\| \sum_i x_i x_i^* \right\| \leq \sum_i \|x_i\|^2 \leq n \left\| \sum_i x_i^* x_i \right\|.$$

Thus $\gamma_n(E) \leq \sqrt{n}$. In $E = \mathbf{R}$, take the first n canonical row vectors $e_i = E_{1i}$. Their column norm is 1, while their row norm is $\sqrt{n}$. Therefore

$$\gamma_n(\mathbf{R}) = \sqrt{n}. \quad (4.2)$$

For $E = \mathbf{C}$, regard x_i as Hilbert-space columns. The row is their synthesis operator, whose operator norm is bounded by its Hilbert–Schmidt norm; the latter is exactly the norm of the stacked column. Hence $\gamma_n(\mathbf{C}) \leq 1$. Equality follows by taking only $x_1 \neq 0$, so

$$\gamma_n(\mathbf{C}) = 1. \quad (4.3)$$

Finally, $\mathbf{OH}$ is completely isometric to its opposite through the identity. Equivalently, transposition preserves the norm of every $\mathbf{OH}$ -valued matrix. (For a column this also follows directly from the usual Gram-matrix formula $\|[x_i]^t\| = (\sum_{i,j} |\langle x_i, x_j \rangle|^2)^{1/4}$.) Consequently,

$$\gamma_n(\mathbf{OH}) = 1. \quad (4.4)$$

Interpolation upper bounds

If $0 \leq \theta \leq \frac{1}{2}$, reiteration gives $H_\theta = (\mathbf{R}, \mathbf{OH})_{2\theta}$. Apply the interpolation theorem to the same transpose map t_n , using (4.1), (4.2), and (4.4):

$$\gamma_n(H_\theta) \leq (\sqrt{n})^{1-2\theta} 1^{2\theta} = n^{(1-2\theta)/2}. \quad (4.5)$$

If $\frac{1}{2} \leq \theta \leq 1$, then $H_\theta = (\mathbf{OH}, \mathbf{C})_{2\theta-1}$, and (4.3)–(4.4) give

$$\gamma_n(H_\theta) \leq 1. \quad (4.6)$$

The exact coordinate witness

It remains to prove equality in (4.5). Work first in the n -dimensional coordinate spaces $\mathbf{R}_n, \mathbf{C}_n$; their inclusions and coordinate projections are complete contractions at both endpoints, so interpolation embeds their interpolants completely isometrically in H_θ .

Let $e_1, \dots, e_n$ be the common canonical basis and set

$$u = [e_1 \ \cdots \ e_n]^t, \quad v = [e_1 \ \cdots \ e_n].$$

The concrete endpoint identifications give

$$\begin{array}{c|cc} & \mathbf{R}_n & \mathbf{C}_n \\ \hline \|u\| & \|I_n\|_{M_n} = 1 & \sqrt{n} \\ \|v\| & \sqrt{n} & \|I_n\|_{M_n} = 1. \end{array} \quad (4.7)$$

Log-convexity of interpolation norms therefore yields

$$\|u\|_{M_{n,1}(H_\theta)} \leq n^{\theta/2}, \quad \|v\|_{M_{1,n}(H_\theta)} \leq n^{(1-\theta)/2}. \quad (4.8)$$

For completeness, both inequalities are equalities. Define the common coordinate functional

$$\varphi([x_i]) = \sum_{i=1}^n (x_i)_i.$$

On the two endpoint realizations of the column u , this is respectively the trace functional on M_n and the inner product with the vector having n unit coordinates in $\ell_2^{n^2}$. Its functional norms are therefore n and $\sqrt{n}$. Applying the interpolation theorem directly to the compatible functional φ gives

$$\|\varphi\|_{(M_{n,1}(H_\theta))^*} \leq n^{1-\theta}(\sqrt{n})^\theta = n^{1-\theta/2}.$$

Since $\varphi(u) = n$, this forces $\|u\| \geq n^{\theta/2}$, proving equality in the first part of (4.8).

For the row v , the same functional has endpoint dual norms $\sqrt{n}$ and n , in that order. Hence

$$\|\varphi\|_{(M_{1,n}(H_\theta))^*} \leq (\sqrt{n})^{1-\theta} n^\theta = n^{(1+\theta)/2}.$$

The identity $\varphi(v) = n$ now gives $\|v\| \geq n^{(1-\theta)/2}$, proving equality in the second part of (4.8). Thus

$$\gamma_n(H_\theta) \geq \frac{\|v\|}{\|u\|} = n^{(1-2\theta)/2} \quad (0 \leq \theta \leq \tfrac{1}{2}).$$

Together with (4.5), this is equality. Above the midpoint, (4.6) and a family with only one nonzero member give $\gamma_n(H_\theta) = 1$. This proves Theorem 2.

Since H_θ^{op} is completely isometric to $H_{1-\theta}$, the row-to-column norm on H_θ is the column-to-row norm on $H_{1-\theta}$. Corollary 3 follows.

5 A general interpolation consequence

Proposition 4 (Interpolation stability). *Let (E_0, E_1) be a compatible couple of operator spaces. If E_j has column-row constant K_j for $j = 0, 1$, then*

$$E_s = (E_0, E_1)_s \quad \text{has column-row constant at most} \quad K_0^{1-s} K_1^s.$$

More precisely, $\gamma_n(E_s) \leq \gamma_n(E_0)^{1-s} \gamma_n(E_1)^s$ for every n .

Proof. By (4.1), the transpose t_n acts between the interpolated column and row couples. The ordinary interpolation theorem for operators gives the displayed finite-level estimate. Taking the supremum in n gives the uniform assertion. $\square$

6 Verification, scope, and novelty limits

The proof has three independent checks: endpoint norms are computed in concrete matrix models; the upper estimate follows from exact interpolation; and the coordinate/trace witness supplies a matching lower estimate. The formulas recover all source-paper endpoints:

$$\theta = 0 : \gamma_n = \sqrt{n}, \quad \theta = \frac{1}{2} : \gamma_n = 1, \quad \theta = 1 : \gamma_n = 1.$$

Opposite-space symmetry reproduces the reverse formula and is an additional consistency check.

This theorem completely classifies the row–column interpolation scale, but the source question ranges over all operator spaces and remains much broader. A bounded search of arXiv and web indexes on 17 August 2026 found the source paper and the complete-Pick-space literature, but no statement matching the exact formula above. That is evidence only, not a definitive novelty claim. Independent expert review should focus on the standard interpolation identifications in (4.1) and on the endpoint norms of the trace witness.

References

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